Khoa Nguyen - Lead Data Engineer

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Lead Data Engineer and former Technical Lead with 10+ years of experience in database architecture, data platforms, and large-scale data systems.

Strong background in SQL-based data modeling, database performance optimization, and scalable data services design. Experienced in evolving legacy systems into modern data architectures and designing and operating data platforms that support both internal analytics and customer-facing data products.

Hands-on experience with relational databases, schema design, query optimization, and distributed data processing systems.

Regularly write technical articles on Medium, sharing guides on software engineering, data-related topics, and personal experiences from real-world projects.

CORE TECHNOLOGIES & EXPERTISE

Engineering Foundations

- Programming: Python, SQL, C#, Go, Java
- Service development: REST APIs, microservices, service architecture
- Data access & ORM: SQL, Entity Framework, Dapper, GORM,
- System design & distributed systems
- Performance tuning, debugging, profiling

Cloud Platforms

- AWS data platform (S3, Glue, Athena, Redshift, Spark workloads)
- Google Cloud data platform (BigQuery)
- Azure data ecosystem (ADLS Gen2, Data Factory, Databricks, Synapse)

Data Platforms & Databases

- Amazon Redshift
- BigQuery
- Snowflake
- ClickHouse
- PostgreSQL, MySQL, SQL Server
- Storage design, partitioning, and retention strategies
- Query performance tuning & workload optimization

Data Orchestration & Transformation

- Airflow, Dagster
- dbt, SQLMesh
- Modular ELT workflows
- Idempotent pipelines and backfill strategies

Data Engineering & Architecture

- ETL / ELT pipeline design (Python-based)
- Event-based & behavioral data modeling
- Streaming + batch hybrid architectures
- Data quality frameworks, validation pipelines, and data reliability

Distributed Data Processing

- Apache Spark
- Databricks / Spark ecosystem
- High-volume batch & near real-time processing
- Low-latency analytics pipelines
- Distributed data processing & performance optimization

Analytics Engineering

- Dimensional modeling & governed semantic layer
- Production-grade dbt modeling
- Funnel, cohort, attribution & exploratory analysis
- Dashboard design & BI platforms (**Metabase, Looker, Holistics**)
- Self-service analytics enablement (**~80% adoption**)

Technical Leadership

- End-to-end platform ownership
- Architecture & technical decision-making
- Mentoring engineers & analysts
- Cross-functional collaboration (Product, Marketing, Finance)
- Engineering best practices (code review, modular design, PR workflows)

PROFESSIONAL & ENGLISH SKILLS

- Fluent English; extensive experience in international working environments
- Team leadership, mentoring, and technical alignment
- Effective collaboration in multicultural and distributed teams
- Strong stakeholder communication and technical facilitation

PROFESSIONAL EXPERIENCES

Lead Data Engineer, Flodesk

Mar 2020 — Dec 2025

Flodesk is a design-focused email marketing platform enabling small businesses to grow through **email-driven marketing**.

- Served as technical lead and end-to-end owner of the company's analytics and data platform, driving architectural decisions and platform evolution.
- Built and operated **large-scale data pipelines** processing 1**00B+ email and user events**, enabling large-scale product and behavioral analytics while supporting both batch and near real-time workloads.
- Enabled **self-service analytics** adoption (**~80% active usage**) by delivering standardized semantic models and analytics-ready datasets used across product, marketing, and finance teams.
- Improved **marketing effectiveness by 30%+** through unified reporting, reliable attribution models, and trusted performance metrics.
- Partnered with Product, Marketing, and Finance stakeholders to define KPIs, analyze funnel and cohort performance, evaluate campaign attribution, and design stakeholder-facing dashboards (Metabase, Holistics, Looker-style BI) that delivered actionable business insights.
- Powered **~20% of customer-facing analytics features** using low-latency, high-availability data systems.
- Defined modular data domains and scalable data services architecture supporting analytics and product-facing workloads.
- Designed and evolved a company-wide data warehouse using modular, model-driven transformation workflows (dbt-based, SQLMesh-aligned patterns) with strict idempotency discipline and re-runnable backfill strategies.
- Architected Medallion-style data layers (raw → curated → business-ready) to ensure scalable transformations and consistent metric definitions across teams.
- Implemented rigorous data validation processes including row-count reconciliation, metric diffing, and controlled parallel runs prior to production rollout to ensure trusted analytics data.
- Designed partition-aware merge strategies and scalable compute/storage separation patterns used in modern cloud data platforms (Redshift, Snowflake, BigQuery, Databricks-based architectures).
- Led and mentored a cross-functional data team, promoting clean SQL standards, modular data modeling practices, and engineering best practices.
- Conducted a Snowflake proof-of-concept to evaluate performance isolation, scalability, and long-term warehouse evolution strategy.

Technical Lead / Software Designer, Kamora Limited Company

Feb 2019 — Feb 2020

- Acted as Technical Lead & key contributor in delivering end-to-end product features across backend, frontend, and mobile platforms.
- Led system design and architecture decisions for multiple startup products targeting European markets, ensuring scalability and production readiness.
- Took ownership of core modules, mentored team members, and drove technical alignment in fast-paced, early-stage environments.
- Ensured stable releases and rapid iteration cycles, balancing speed and quality for client-facing European projects.

Technical Lead / Software Designer, Sioux High Tech Software

Aug 2013 — Jan 2019

- Served as Technical Lead / key member in the design and delivery of large-scale, mission-critical systems for European enterprise and government clients.
- Led end-to-end system design, from requirements analysis to architecture definition and production deployment.
- Worked directly with European customers on technical requirements, architectural trade-offs, and long-term system evolution.
- Played a central role in delivering systems with high reliability, safety, and compliance standards.
- Collaborated and led within multinational engineering teams, maintaining high technical quality and best practices across distributed teams.

HIGHLIGHTED PROJECTS

Company-wide Data Platform & Single Source of Truth, United States

- Built a unified data foundation consolidating **product, marketing, and finance data** into a consistent analytics layer.
- Established standardized datasets and metrics to ensure **data consistency and trust** across business functions.
- Enabled organization-wide analytics adoption, supporting **self-service reporting for ~80% of frequent users**.

High-volume Event Analytics for Email Marketing, United States

- Designed a **high-throughput event ingestion and analytics system** handling **100B+ email and user events**.
- Supported both **near real-time and batch analytics** for data exploration and customer-facing use cases.
- Provided a reliable analytics backbone for **~20% of customer-facing product features**.

Self-service Analytics Enablement, United States

- Developed analytics models and BI-ready datasets to support **self-service analysis across business teams**.
- Reduced reliance on ad-hoc data requests by enabling users to independently explore and answer business questions.
- Improved overall **data literacy and analytics efficiency** across the organization.

Smart Car, The Netherlands

- Contributed to a **government-backed initiative in the Netherlands** aimed at reducing traffic congestion on **highway A58**.
- Built a system to **collect and process real-time vehicle telemetry data via USB** from in-car devices.
- Analyzed driving patterns and traffic conditions to generate **driver-facing instructions and speed recommendations**.
- Supported improved traffic flow through **real-time, data-driven guidance** delivered on mobile devices.

ACHIEVEMENTS

- Microsoft Certified Xamarin Developer
- Speaker at Vietnam Mobile Day

EDUCATION

Bachelor of Information Technology (Excellent)

Da Nang University of Technology, 2008 — 2013

Additional Information

Hobbies: Football, Writing technical articles